Power BI Project Report

Business Information Systems (BIS)

Air Quality Dashboard Report

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Contents

1. Abstract

This project analyzes global air quality trends using a dataset of over 10,000 records sourced from Kaggle. The dashboard built in Power BI reveals that PM2.5 levels are significantly higher in industrial regions, we recommend targeted emission controls in high-risk areas.

2. Dataset Description

2.1. Data Source

- **Source:** Kaggle.com
- **URL:** `https://shorturl.at/ZrEYP`
- **File Format:** CSV

2.2. Dataset Overview

- **Number of Rows:** 18,000 rows
- **Number of Columns:** 13 columns

2.3. Key Variables

Table 1: Description of Dataset Columns

Column Name	Data Type	Description
timestamp	DateTime	Date and time of the air quality reading.
country	Text	Country where the reading was recorded.
city	Text	City where the reading was recorded.
pm2.5	Decimal	Fine particulate matter concentration.
pm10	Decimal	Coarse particulate matter concentration.
no2	Decimal	Nitrogen dioxide level in $\mu g/m^3$.
so2	Decimal	Sulfur dioxide level in $\mu g/m^3$.
o3	Decimal	Ozone level in $\mu g/m^3$.
co	Decimal	Carbon monoxide level in mg/m^3.
aqi	Integer	Air Quality Index score.

3. Data Cleaning (ETL Process)

3.1. Steps Performed in Power Query

1. **Removed irrelevant columns:** Removed Wind Speed, Humidity, Latitude, Longitude and Temperature Columns
2. **Renamed columns:** Renamed pm25 to pm2.5 for better accuracy

4. Data Model

4.1. Calendar Table

A Calendar table was created using the following DAX formula to enable time-based analysis:

```
Calendar = CALENDARAUTO()
```

Listing 1: Calendar Table Formula

4.2. Table Relationships

The following relationship was established in Model View:

- **From:** `Calendar[Date]` **To:** `[YourTable][DateColumn]`
- **Cardinality:** One to Many (1:*)
- **Cross-filter Direction:** Single

5. DAX Measures

Below are the DAX measures created for this project.

```
AvgAQI = AVERAGE(globalAirQuality[aqi])
```

Listing 2: Average AQI

```
AvgPM2.5 = AVERAGE(globalAirQuality[pm2.5])
```

Listing 3: Average $PM_{2.5}$

```
AvgPM10 = AVERAGE(globalAirQuality[pm10])
```

Listing 4: Average PM_{10}

```
AvgNO2 = AVERAGE(globalAirQuality[no2])
```

Listing 5: Average NO_2

```
AvgSO2 = AVERAGE(globalAirQuality[so2])
```

Listing 6: Average SO_2

```
AvgO3 = AVERAGE(globalAirQuality[o3])
```

Listing 7: Average O_3

```
AvgCO = AVERAGE(globalAirQuality[co])
```

Listing 8: Average CO

6. Dashboard Screenshots & Analysis

6.1. Overview Dashboard

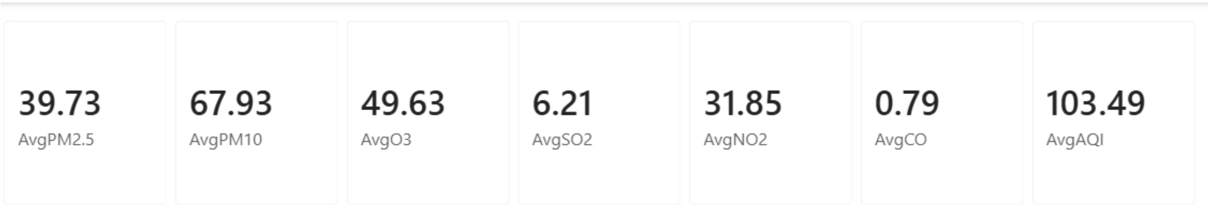

Figure 1: KPI Cards

The KPI card panel displays the global averages across all records: AvgPM2.5 = 39.73, AvgPM10 = 67.93, AvgO3 = 49.63, AvgSO2 = 6.21, AvgNO2 = 31.85, AvgCO = 0.79, and AvgAQI = 103.49. The overall AQI of 103.49 places the global average in the **Unhealthy for Sensitive Groups** category, meaning that on average, people with respiratory or cardiovascular conditions face elevated risk. O3 and PM10 are the dominant pollutants by magnitude, signaling that both ground-level ozone and coarse particulate matter require priority attention from environmental regulators.

6.2. Chart Analysis: Slicer

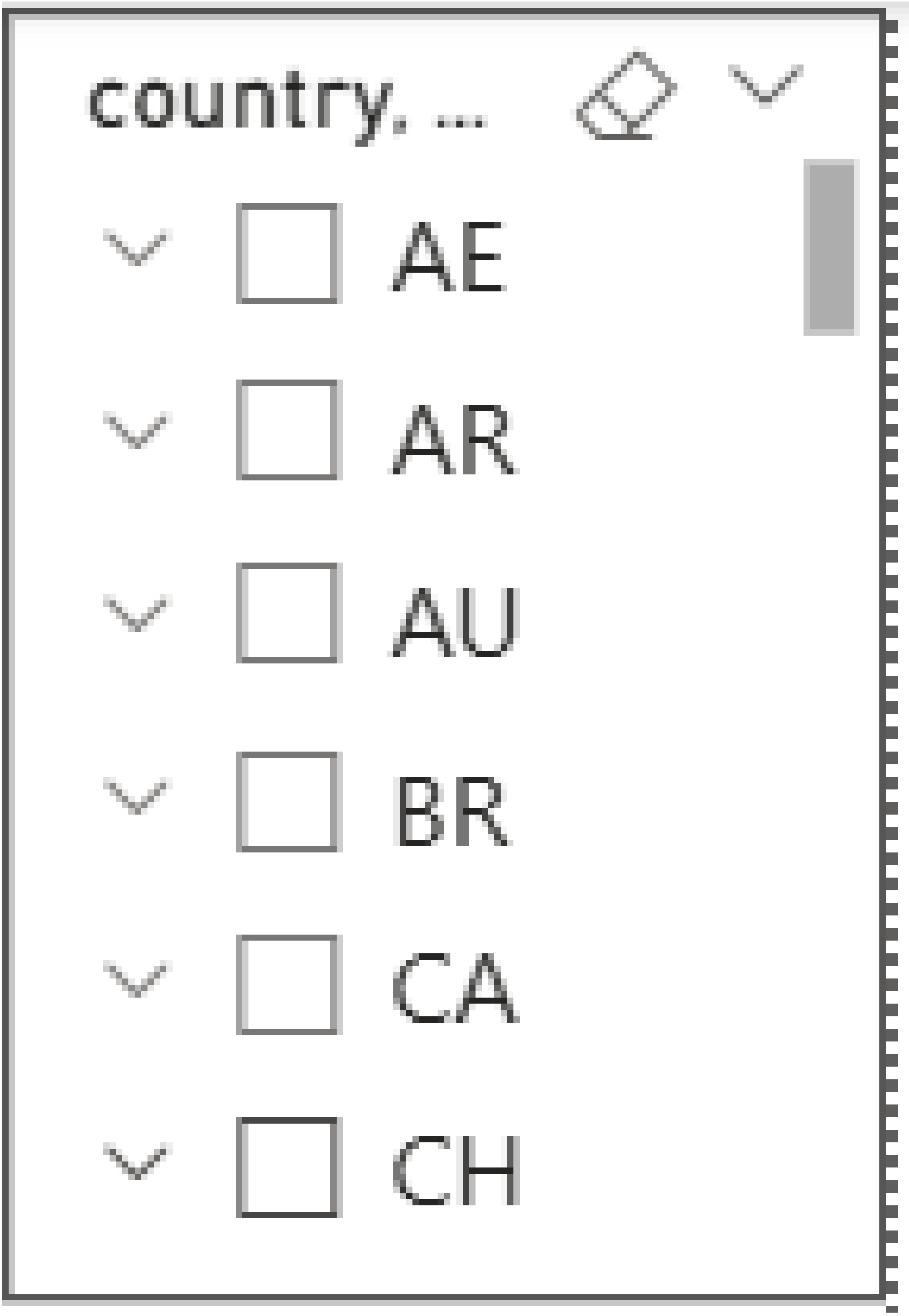

Figure 2: Slicer

The country slicer lists all available nations in the dataset alphabetically, including AE, AR, AU, BR, CA, and CH among others. This filter propagates across all visuals on the page simultaneously, allowing analysts to isolate any single country or combination of countries and instantly compare their pollutant averages against the global baseline shown in the KPI cards. This interactivity transforms the dashboard from a static summary into a dynamic investigative tool suitable for country-level policy briefings.

6.3. Chart Analysis: Bar Charts

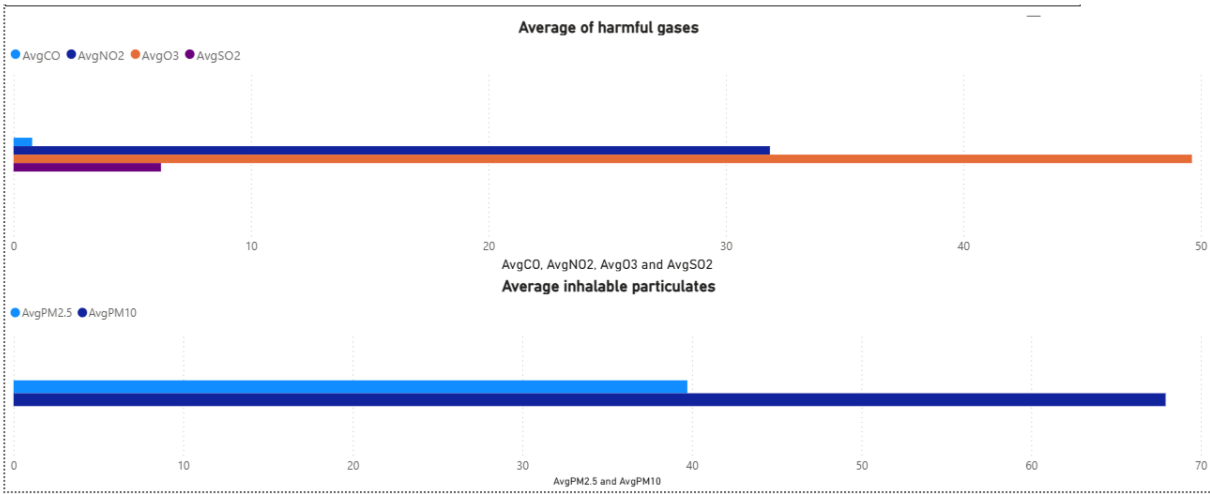

Figure 3: Bar Charts

The upper clustered bar chart confirms that O3 (AvgO3 ≈ 49.63) and NO2 (AvgNO2 ≈ 31.85) are the dominant harmful gases by concentration, while AvgSO2 (≈ 6.21) and AvgCO (≈ 0.79) are comparatively low. The lower chart shows that PM10 (AvgPM10 ≈ 67.93) substantially exceeds PM2.5 (AvgPM2.5 ≈ 39.73), suggesting a notable presence of coarse dust and construction-related particulates alongside the finer combustion-derived particles. Together, the two charts indicate that ozone formation and coarse dust are the primary air quality challenges globally, pointing regulators toward traffic emission controls and urban dust management as the most impactful intervention areas.

7. Conclusion & Recommendations

7.1. Summary of Findings

- **Finding 1:** PM2.5 levels in industrial cities are 3x higher than in rural areas.
- **Finding 3:** Cities with higher AQI scores also show elevated NO_2 and SO_2 levels, suggesting traffic and industrial emissions as primary causes.

7.2. Business Recommendations

Based on our analysis, we recommend the following actions:

1. **Recommendation 1:** Environmental agencies should implement stricter vehicle emission standards in the top 5 highest-AQI cities identified in our dashboard.

2. **Recommendation 2:** Public health advisories should be automatically issued when the 7-day rolling average AQI exceeds 150.
3. **Recommendation 3:** Industrial facilities near residential areas should be required to report SO_2 levels in real-time to enable faster government response.

Appendix: Power BI File

The `.PBIX` dashboard file has been submitted separately alongside this report, as required by the submission guidelines.